

Math check

Name: _____ Date: _____

This is not a pass/fail test. It shows what you already know, so we know where to start. Each section starts easy and gets harder. Nobody is expected to do everything. If you get stuck, write down what you would try, then move on.

Working: do your working on the lined paper, numbered by question. Write only the final answer on this sheet.

Calculator: not allowed in Sections 1 and 2. Allowed in Sections 3, 4 and 5.

After each answer, circle how sure you are: **1** = guessing, **2** = fairly sure, **3** = certain.

What a good answer looks like, everywhere:

1. Working that someone else can follow, one step per line.
2. The final answer clearly marked, with units (\$, cm, °) where needed.
3. A quick check: does the answer make sense? (A sale price should be lower than the original price.)

Before you start: how long do you think this will take? _____ minutes. Start time: _____

Section 1: Number

1.1 A jacket costs \$80. In a sale, the price is reduced by 15%. What is the sale price?

What to do: find how much 15% of \$80 is, then work out what the jacket costs after that amount is taken off.

Good answer: shows the amount taken off and the final price in \$.

Answer: _____

Sure? 1 2 3

1.2 (a) Write 3.2×10^4 as an ordinary number. (b) Work out $(3 \times 10^5) \times (4 \times 10^{-2})$. Give the answer in scientific notation.

What to do: "scientific notation" means a number between 1 and 10, multiplied by a power of 10 (for example $4.7 \times 10^3 = 4700$). In (b), multiply, then make sure your answer is in that form.

Good answer: (a) a normal number with all its zeros; (b) one number between 1 and 10×10 to some power.

(a) _____ (b) _____

Sure? 1 2 3

1.3 A colony of 2000 bacteria grows by 10% every hour. How many bacteria are there after 3 hours?

What to do: each hour, the number goes up by 10% of whatever is there at that moment, not 10% of the starting 2000.

Good answer: shows the number after hour 1, hour 2 and hour 3 (or one calculation that does all three at once).

Answer: _____

Sure? 1 2 3

1.4 Simplify $\sqrt{50} + \sqrt{8}$.

What to do: the answer should still contain a square root, written as simply as possible (a whole number $\times \sqrt{\text{of a small number}}$). Do not turn it into a decimal.

Good answer: a single term like $5\sqrt{3}$, with the steps that got you there.

Answer: _____

Sure? 1 2 3

Section 2: Algebra

2.1 Solve $5(x - 2) = 3x + 4$.

What to do: "solve" means find the value of x that makes both sides equal. Deal with the parentheses first.

Good answer: $x =$ a number, and a check: put your number back in and show both sides match.

$x =$ _____

Sure? 1 2 3

2.2 A straight line passes through the points (1, 3) and (4, 9). Find its equation.

What to do: write it as $y = mx + b$, where m is the slope (how much y goes up when x goes up by 1) and b is where the line crosses the y -axis.

Good answer: the slope worked out and shown, then b , then the full equation, checked with both points.

$y =$ _____

Sure? 1 2 3

2.3 Solve these two equations together: $2x + y = 11$ and $x - y = 1$.

What to do: find one value of x and one value of y that make **both** equations true at the same time.

Good answer: $x = \dots$, $y = \dots$, and a check in both equations.

$x =$ _____ $y =$ _____

Sure? 1 2 3

2.4 Solve $x^2 + 4x - 3 = 0$.

What to do: find every value of x that makes the left side equal 0. This one does not factor nicely, so you need a method that always works. Leave square roots in the answer (no calculator).

Good answer: both solutions in exact form (with $\sqrt{\quad}$), the method written out step by step.

$x =$ _____

Sure? 1 2 3

Section 3: Geometry (calculator allowed)

3.1 A circle has a radius of 5 cm. Find (a) its area and (b) its circumference.

What to do: radius = distance from the centre to the edge. Circumference = the distance all the way round. Use π (the π button, or 3.14).

Good answer: the formula written first, then the numbers, then the answer rounded to 1 decimal place with the right units (cm^2 for area, cm for length).

(a) _____ (b) _____

Sure? 1 2 3

3.2 A 5 m ladder leans against a vertical wall. Its foot is 1.4 m from the wall. How high up the wall does it reach?

What to do: draw it first. The wall, the ground and the ladder make a right triangle. You know two sides, so find the third.

Good answer: a labelled sketch, the method, and the answer in m . Check: it must be less than 5 m . Why?

Height: _____

Sure? 1 2 3

3.3 A 1.8 m tall person casts a 2.4 m shadow. At the same moment, a tree casts a 16 m shadow. How tall is the tree?

What to do: sketch both. The sun hits both at the same angle, so the two triangles have the same shape (one is a scaled-up copy of the other).

Good answer: the sketch, the scale factor or ratio shown, and the height in m .

Height: _____

Sure? 1 2 3

3.4 A right triangle has a longest side (hypotenuse) of 12 cm. One of the other angles is 35° . Find the length of the side opposite the 35° angle.

What to do: sketch it and label the sides you know and the side you want. This needs $\sin$, $\cos$ or $\tan$. Choose the one that links those two sides.

Good answer: the sketch, which ratio you chose and why, and the answer to 1 decimal place in cm .

Length: _____

Sure? 1 2 3

Section 4: Data and probability (calculator allowed)

4.1 For the numbers 4, 7, 7, 9, 13, find the mean, the median and the range.

What to do: mean = add them up and share equally. Median = the middle one when in order. Range = biggest minus smallest.

Good answer: all three, each labelled.

Mean _____ Median _____ Range _____

Sure? 1 2 3

4.2 A bag has 3 red, 5 blue and 2 green counters. One is picked at random. (a) What is the probability it is not blue? (b) If you pick a counter 50 times, putting it back each time, about how many reds would you expect?

What to do: probability = number of ways to get what you want ÷ total number of possibilities.

Good answer: (a) a fraction in its simplest form; (b) a number, with one line saying how you got it.

(a) _____ (b) _____

Sure? 1 2 3

4.3 The table shows goals scored by a team in 20 matches. Find the mean number of goals per match.

Goals	0	1	2	3
Matches	4	6	7	3

What to do: the table means, for example, the team scored 1 goal in 6 different matches. Find the total number of goals across all 20 matches first.

Good answer: the total goals shown, then the mean as a decimal.

Mean: _____

Sure? 1 2 3

4.4 In a school of 200 students, 120 play a sport. Of those, 45 also play an instrument. Of the students who do not play a sport, 30 play an instrument. A student is picked at random and you are told they play an instrument. What is the probability they also play a sport?

What to do: a two-way table (sport / no sport × instrument / no instrument) helps. Being told "they play an instrument" changes which group you are picking from.

Good answer: the completed table, the group you are picking from named, and the probability as a fraction or decimal.

Answer: _____

Sure? 1 2 3

Section 5: Think it through

Two phone plans: Plan A costs \$15 a month, unlimited calls. Plan B costs \$5 a month plus 10¢ per minute of calls.

(a) When is Plan A cheaper, and when is Plan B cheaper? Find the exact point where they cost the same.

(b) Which would you choose for yourself, and why?

(c) What else would you want to know before deciding for real?

What to do: before you start, write 1–2 lines on how you plan to tackle it. Use any method: a table, a graph, an equation, or words.

Good answer: the break-even point found and explained, a choice backed by a reason, and at least two sensible things you would still want to know.

When you finish

End time: _____

Which section felt easiest? _____ Hardest?

Did you change your mind halfway through any question? Which one, and why?